Yifu Luo

(+49) 1759402517 yifu599@gmail.com Homepage

EDUCATION

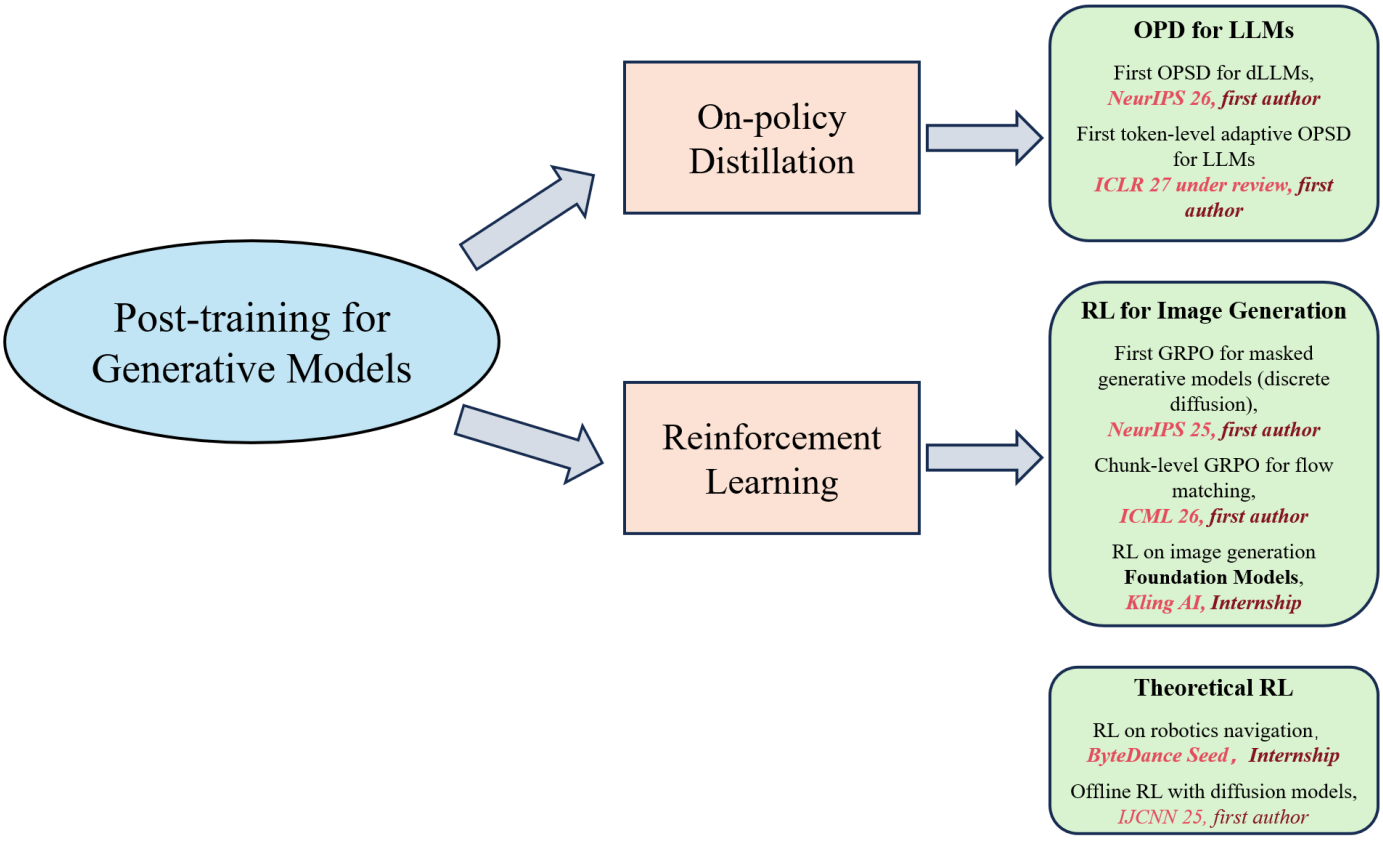
Ph.D., Technical University of Munich, Germany *Oct 2026 - Sep 2029*
Computer Science.
Advised by Prof. Dr. Gjergji Kasneci.

M.E., Tsinghua University, China *Sep 2023 - Jun 2026*
Artificial Intelligence Program, Electronic and Information Engineering.
Advised by Prof. Dr. Xueqian Wang.

B.E., Tsinghua University, China *Aug 2018 - Jun 2022*
Electronics Information Engineering

RESEARCH INTEREST

I am interested in advancing the post-training paradigms of generative models, with a particular emphasis on reinforcement learning (RL) and on-policy distillation (OPD), for LLMs, diffusion models, and agentic systems. My past works range from RL post-training to enhance diffusion models (image generation) to self-OPD on LLMs. I have also explored theoretical RL and its applications on robotics.



INDUSTRY RESEARCH EXPERIENCE

Research Intern, Kling AI *Jul 2025 - Oct 2025*
Mentored by Dr. Changqian Yu
– I enhanced flow-matching text-to-image models (FLUX) through RL post-training. I shifted GRPO from the step-level to the chunk-level yielding superior image generation performance. *(**ICML 26, first author**)*
– I also developed reward models using Qwen-VL and InternVL for image generation **foundation models**, including the data construction and model training.

Research Intern, ByteDance Seed *Mar 2025 - Jul 2025*
Mentored by Dr. Yansheng Wang
– I developed RL and imitation learning methods for robotic decision-making from video generation.
– I also developed reward models using Qwen-VL for robotics spatial reasoning and decision-making.

ACADEMIC RESEARCH EXPERIENCE

Research Assistant, Technical University of Munich *Oct 2026 - Sept 2029*

Advised by Prof. Dr. Gjergji Kasneci

– I identified that even within the same response, different tokens prefer different teachers. Based on this, I proposed the first token-level adaptive OPSD for LLMs, constructing a multi-teacher pool and adaptively assigning each token its own teacher. Our approach yields superior performance in **math reasoning tasks**. *(**ICLR 27 under review, first author**)*

Research Intern, Max Planck Institute for Intelligent Systems *Jan 2026 - Jun 2026*

Advised by PI. Shiwei Liu

– I proposed the first On-policy Self-distillation (OPSD) approach for diffusion Large Language Models (dLLMs). We shifted the OPSD formulation from prefix conditioning to self-suffix generation conditioning. Our approach yields superior performance in **math reasoning tasks**. *(**NeurIPS 26, first author**)*

Research Assistant, Tsinghua University *Sep 2023 - Jun 2026*

Advised by Prof. Xueqian Wang

– I focused on RL and its post-training paradigms to enhance generative models.

– I proposed the first GRPO approach for masked generative models in text-to-image generation, with a novel transition probability tailored for them. *(**NeurIPS 25, first author**)*

– I also proposed a frequency-aware decision diffuser for offline RL. *(**IJCNN 25, first author**)*

PUBLICATIONS [Google Scholar] ★ Highlight

Accepted

★ [1] **Yifu Luo**, Zeyu Chen, et al. "Learning from the Self-future: On-policy Self-distillation for dLLMs". In *Neural Information Processing Systems (**NeurIPS**)*, 2026. [arxiv][code]

– Propose the first OPSD approach for dLLMs. We reframe the OPSD formulation from prefix conditioning to the self-suffix generation conditioning tailored specifically for dLLMs. We also shift dense supervision from the token-level to step-level, aligning training with the iterative denoising nature of dLLMs.

★ [2] **Yifu Luo**, Xinhao Hu, et al. "Reinforcement Learning Meets Masked Generative Models: Mask-GRPO for Text-to-Image Generation". In *Neural Information Processing Systems (**NeurIPS**)*, 2025. [arxiv][code]

– Propose the first GRPO approach for masked generative models in text-to-image generation. We redefine the transition probability tailored specifically for masked generative models, and explore several useful strategies to further enhance our method.

★ [3] **Yifu Luo**, Haoyuan Sun, et al. "Principled RL for Flow Matching Emerges from the Chunk-level Policy Optimization". In *International Conference on Machine Learning (**ICML**)*, 2026. [arxiv][code]

– Propose a chunk-level RL optimization for flow-matching text-to-image generation. We shift GRPO from the step-level to the chunk-level, aggregating consecutive steps into a coherent "chunk" to mitigate the sparse rewards bottleneck.

[4] **Yifu Luo**, Yongzhe Chang, et al. "Wavelet Fourier Diffuser: Frequency-Aware Diffusion Model for Reinforcement Learning". In *International Joint Conference on Neural Networks (**IJCNN**)*, 2025. [arxiv]

– Propose a frequency-aware decision diffuser for offline RL. We introduced frequency analysis into diffusion-based decision making for superior stability.

[5] Haoyuan Sun, Bo Xia, **Yifu Luo**, et al. "Calibration Enhanced Decision Maker: Towards Trustworthy Sequential Decision-Making with Large Sequence Models". In *Transactions on Machine Learning Research (**TMLR**)*. [Openreview]

– Propose a decision-making method for minimizing the state-action marginal distribution distance and enhancing the agent's calibration.

[6] Bo Xia, **Yifu Luo**, et al. "D3D: Conditional Diffusion Model for Decision-Making Under Random Frame Dropping". In *International Conference on Robot and Human Interactive Communication (**ROMAN**)*,2024. [IEEE]

– Propose a conditional diffusion method for decision making under random frame dropping.

[7] Sinan Du, Jiahao Guo, Bo Li, Shuhao Cui, Zhengzhuo Xu, **Yifu Luo**, et al. "VQRAE: Representation Quantization Autoencoders for Multimodal Understanding, Generation and Reconstruction". In *Computer Vision and Pattern Recognition (**CVPR**)*, 2026. [arxiv][code]

– Propose a quantization version of autoencoders for unified visual tasks for both understanding and generation.

[8] Haoxiang Cao, Yuxuan Zhang, Penghui Du, Bo Li, Jiajiong Cao, Yingying Fan, Zimeng Wu, **Yifu Luo**, et al. "LongBanana: An Expert-Verified Benchmark for Long-Context Multi-Reference Image Synthesis". In *Neural Information Processing Systems (**NeurIPS**) ED Track*, 2026.

– We introduce the first expert-designed and expert-verified benchmark for long-context multi-reference image synthesis.

Under Review

★ [1] **Yifu Luo**, Chuang Ma, et al. “Every Token Needs Its Own Teacher: Multi-Teacher Adaptive On-Policy Self-Distillation”. Under review at *International Conference on Learning Representations (**ICLR**)*, 2027. [arxiv][code]
– Identified that even within the same response, different tokens prefer different teachers. Based on this, we proposed the first token-level adaptive OPSD for LLMs, constructing a multi-teacher pool and adaptively assigning each token its own teacher.

[2] Shenhui Zhang, Penghui Du, Bo Li, Lai jiang, Haoxiang Cao, **Yifu Luo**, et al. “DuetGRPO: Co-Adaptive Joint Reinforcement Learning for Reasoning-Driven Image Generation”. Under review at *International Conference on Learning Representations (**ICLR**)*, 2027.
– Proposed a joint RL framework for reasoning-driven image generation, with empirical insights into reward assignment and joint optimization between text supervision and image-text alignment.

[3] Xu Yang, Zhizhou Sha, Junbo Li, Jian Yu, Yifan Sun, Matthew Zhao, Jinrui Fang, Xinyue Guo, Yining Wu, Xu Hu, **Yifu Luo**, et al. “No Hidden Prompts Needed! You Can Game AI Peer Review with Presentation-Only Revisions”. Under review at *Annual Meeting of the Association for Computational Linguistics (**EACL**)*, 2027. [arxiv][code]
– An interesting work on AI-generated reviews, with analysis insights revealed, a benchmark, and a framework.

Preprint

[1] Haoyuan Sun, Jing Wang, Yuxin Song, Yu Lu, Bo Fang, **Yifu Luo**, et al. “Power Reinforcement Post-Training of Text-to-Image Models with Super-Linear Advantage Shaping”. [arxiv]
– Proposed Super-Linear Advantage Shaping (SLAS) to mitigate the normalization issues in the text-to-image RL post-training.

[2] Haoyuan Sun, Jiaqi Wu, Bo Xia, **Yifu Luo**, et al. “Reinforcement Fine-Tuning Powers Reasoning Capability of Multimodal Large Language Models”. [arxiv]
– Argue reinforcement fine-tuning powers the reasoning capability of multimodal large language models.

LANGUAGE: IELTS 8.0